Simplifying fractions

1 Complete the statement. A factor is a term which exactly divides another term. Fill in the blank

2 Select all the fractions that simplify to $\frac{2}{3}$. Tick **2** correct answers

- [x] $\frac{8}{12}$
- [ ] $\frac{9}{12}$
- [ ] $\frac{15}{18}$
- [x] $\frac{12}{18}$
- [ ] $\frac{9}{15}$

3 Which of the following factors of 24 and 90 is the best choice to efficiently simplify $\frac{24}{90}$?

Tick **1** correct answer

- [ ] 3
- [x] 6
- [ ] 12
- [ ] 15

4 Write 180 as a product of prime factors. Tick **1** correct answer

- [x] $2^2 \times 3^2 \times 5$
- [ ] $2 \times 3^2 \times 5$
- [ ] $2 \times 3^2 \times 5^2$
- [ ] $2 \times 3 \times 5$
- [ ] $2^2 \times 3^2 \times 5^2$

5 Which of these fractions is equivalent to $\frac{360}{700}$ Tick **1** correct answer

- [ ] $\frac{2^2 \times 3^2 \times 5}{2^2 \times 5^2 \times 7}$
- [ ] $\frac{2^3 \times 3^2 \times 5}{2^2 \times 5 \times 7}$
- [ ] $\frac{2^3 \times 3^2 \times 5^2}{2^2 \times 5^2 \times 7^2}$
- [x] $\frac{2^3 \times 3^2 \times 5}{2^2 \times 5^2 \times 7}$
- [ ] $\frac{2^3 \times 3 \times 5}{2^2 \times 5 \times 7}$

6 Match each fraction to its simplified equivalent fraction. Use the fact that $120 = 2^3 \times 3 \times 5; 84 = 2^2 \times 3 \times 7$ and $210 = 2 \times 3 \times 5 \times 7$ Write the correct letter in each box

a	$\frac{84}{210}$
b	$\frac{84}{120}$
c	$\frac{120}{210}$
d	$\frac{120}{84}$

a	$\frac{2}{5}$
d	$1\frac{3}{7}$
b	$\frac{7}{10}$
c	$\frac{4}{7}$